

# Robust Federated Learning Against Adversarial Attacks

*Securing Collaborative Medical AI from Malicious Participants*

**Team : Transformers**

Week 2 — Data Exploration and Baseline

May 2026

## 1. Dataset and Access

This week uses the local copy of the lightweight **CheXpert-v1.0-small** chest X-ray dataset. The official dataset page is:

<https://stanfordmlgroup.github.io/competitions/chexpert/>

CheXpert access requires accepting the dataset terms through Stanford’s official download page. In this repository, the required files are already available under `data/raw/`: the metadata file `train.csv` and the image directory `CheXpert-v1.0-small/train/`. Therefore, no extra credentialing step is needed to reproduce the Week 2 notebooks locally.

The dataset contains 223,414 image rows from 64,540 patients. Among them, 191,027 images are frontal views and 32,387 are lateral views. The project focuses on the five labels already defined in the codebase: Cardiomegaly, Edema, Consolidation, Atelectasis, and Pleural Effusion.

## 2. Exploratory Data Analysis

The metadata contains patient age, sex, view type, projection, image path, and multi-label disease annotations. All image paths in the local metadata were checked and resolved successfully, giving 223,414 existing image rows out of 223,414.

The label columns contain positive values, negative values, uncertain values, and missing values. For this first baseline, the preprocessing policy follows the existing project configuration:

positive = 1,   negative = 0,   uncertain =  $-1 \rightarrow 0$ ,   missing  $\rightarrow 0$ .

After filtering to frontal images and applying this policy, the class distribution is imbalanced:

| Label            | Positive | Negative | Positive Rate |
|------------------|----------|----------|---------------|
| Cardiomegaly     | 23,385   | 167,642  | 12.24%        |
| Edema            | 49,675   | 141,352  | 26.00%        |
| Consolidation    | 12,983   | 178,044  | 6.80%         |
| Atelectasis      | 29,720   | 161,307  | 15.56%        |
| Pleural Effusion | 76,899   | 114,128  | 40.26%        |

The largest missing-label counts in the full CSV occur for Cardiomegaly, Consolidation, and Atelectasis. This is expected for weakly labeled radiology datasets, but it means that evaluation must report per-label metrics and not only global accuracy.

## 3. Preprocessing

The Week 2 preprocessing requirements are:

- use frontal images only for the initial experiments,
- extract patient identifiers from image paths,
- split data by patient to reduce leakage risk,
- map uncertain and missing labels to zero for the first baseline,
- resize images to a fixed resolution,
- normalize images when using CNN-based feature extraction.

The notebook `src/W2_data_exploration.ipynb` implements the EDA and displays plots for demographics, missing labels, class imbalance, and example images. The notebook `src/W2_baseline_experiment` implements a standalone FedAvg baseline directly inside the notebook, without importing existing project source files.

## 4. Baseline Experiment

The first runnable baseline predicts **Cardiomegaly** from chest X-ray images. Because this project focuses on Federated Learning, the baseline is a minimal **Federated Averaging (FedAvg)** experiment implemented directly in the notebook. The reported run uses a larger natural-distribution sample of 10,000 frontal images. Images are converted to grayscale, resized to  $32 \times 32$ , flattened, standardized, and used by a small multilayer perceptron classifier.

The split uses 80% for training and 20% for testing, stratified by the target label. This produces 8,000 training images and 2,000 test images. The training set is divided across five simulated clients. At each communication round, every client receives the global model, trains locally for one epoch, and sends its parameters back to the server. The server computes the weighted average:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k$$

The reported run uses 20 communication rounds.

| Metric            | Value |
|-------------------|-------|
| Accuracy          | 0.878 |
| Balanced Accuracy | 0.511 |
| F1 Score          | 0.054 |
| AUROC             | 0.721 |
| Average Precision | 0.257 |

The accuracy is above 0.8, but it must be interpreted carefully because Cardiomegaly is highly imbalanced in the sampled data and most examples are negative. The low F1 score and balanced accuracy show that accuracy alone is not sufficient for this medical imaging task. However, the experiment verifies that the local data can be loaded, preprocessed, partitioned into clients, trained with FedAvg, and evaluated with standard classification metrics.

## 5. Short Discussion

The EDA confirms that CheXpert is suitable for the project but requires careful preprocessing because of missing labels, uncertain annotations, and strong class imbalance. Pleural Effusion

is the least sparse of the selected labels, making it a reasonable first binary target for debugging the pipeline.

For the next stage, this baseline should be extended incrementally: first by replacing low-resolution pixel features with a stronger frozen CNN feature extractor, then by adding attacked FedAvg and robust aggregation methods such as Krum, coordinate-wise median, and trimmed mean. The Week 2 notebooks provide the foundation for those experiments by validating data access, image loading, client partitioning, FedAvg training, and core metric computation.